



TOMAS URIZAR KLEINKNECHT

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SUMMARY

Industrial Engineer with 3+ years of experience in operations, analytics, and technology development. Skilled in process automation, data science, and software solutions, delivering dashboards, predictive tools, and optimized workflows. Known for analytical thinking, data-driven decision making, and strong stakeholder relationships across cross-functional teams.

WORK EXPERIENCE

Machine Learning & Computer Vision Engineer Mirai Engineering, Germany | Oct 2025 - Present

- Designed and trained a U-Net model for kidney segmentation using PyTorch, achieving 0.88 Dice score on validation data while handling extreme class imbalance.
- Built a full-stack web application (FastAPI + React/TypeScript), enabling batch processing and privacy-preserving clinical deployment via Docker with ~1s/image processing.

Working Holiday - Australia

Australia | Mar 2023 - Jun 2025

- Snowmaker, Selwyn Snow Resort (Season 2024) – Operated and optimized snowmaking systems, ensuring efficient resource use under varying weather conditions.
- Waste Water Treatment Supervisor, Selwyn Snow Resort (Season 2023) – Oversaw end-to-end wastewater treatment operations, sensor calibrations, and coordination of sludge extraction.
- Software Developer – Developed websites, data analysis tools, and automation solutions.

Regional Operations Coordinator

RedSalud CCHC, Chile | Jun 2022 - Dic 2022

- Conducted comprehensive analysis of operational data and patient satisfaction surveys to enhance patient experience.
- Led cross-functional meetings to implement data-driven management systems and technical tools, optimizing staffing allocation across regional clinics.
- Designed and executed organizational chart restructuring, standardizing key roles and improving operational efficiency.

Technical Supervisor

Crystal Lagoons, Chile | Sep 2019 - Dec 2021

- Supervised 6 lagoon and water treatment systems nationwide (~15% of global lagoons), ensuring optimal performance via telemetry, automated dosing, and real-time water quality monitoring.
- Developed automated Salesforce dashboards enabling real-time failure metrics visualization, reducing response time to critical issues and supporting predictive maintenance.
- Streamlined export operations with dashboards, reducing costs by \$3,000 USD per shipment.
- Led client meetings to review metrics, provide technical guidance, and maintain strong operational relationships, ensuring effective communication with 100% of clients.
- Conducted OPEX analysis and technical evaluations for all new prospective clients.
- Implemented innovative lagoon technologies, including floating systems, temperature-controlled zones, GPS tracking and automated energy monitoring.

EDUCATION

Industrial Civil Engineering
Adolfo Ibañez University, Chile.

2014 - 2019

ADDITIONAL INFORMATION

- **Technical Skills:** Python, PyTorch, JavaScript, SQL, APIs, Docker, React, Workflow Automation, Salesforce, ETL Tools, Tableau, Machine Learning, CNN
- **Languages:** Spanish - Native, English - Proficient (C1 - IELTS)